

SQL AI OPERATIONS & ALERTING CENTER

Enterprise-Grade GenAI + SQL Server + WhatsApp

TAQHI MA

4-WEEK HANDS-ON BUILD

From GenAI prototype thinking to an integrated, security-aware operations platform.



Executive Overview

A practical GenAI operations platform

- 01 Natural language database interaction
- 02 SQL Server health monitoring
- 03 WhatsApp operational alerting
- 04 Local Qwen3 4B inference with Ollama
- 05 Streamlit unified operations dashboard
- 06 Application-level read-only SQL guardrails

The Problem

From database complexity to actionable operations

- 01 SQL knowledge is required for many routine questions
- 02 Database health checks are often separated from user workflows
- 03 AI-generated SQL must be constrained before execution
- 04 Operational events benefit from immediate communication
- 05 A useful enterprise assistant needs reliability, security and observability

The Solution

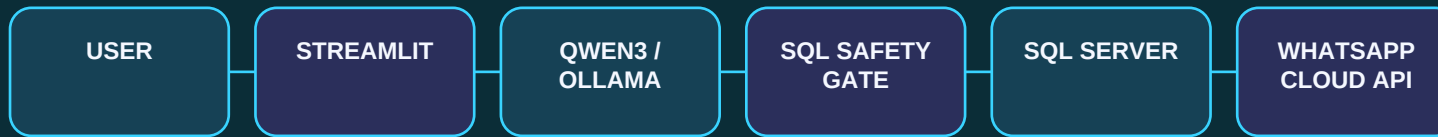
One workflow for AI-assisted database operations

- 01 User → Natural Language
- 02 Qwen3 4B / Ollama → SQL generation
- 03 SQL validation → read-only guardrail
- 04 SQL Server → query execution
- 05 Pandas / Streamlit → results
- 06 WhatsApp Cloud API → operational notification

High-Level Architecture

AI + database + monitoring + messaging

Integrated path: user intent → model → control → data → notification



Core AI Capability

Natural language → SQL → result

- 01 Example: “Show employees with salary greater than 50,000.”
- 02 Qwen3 converts the request into a SQL Server SELECT statement
- 03 Generated SQL is displayed for transparency
- 04 Validation happens before execution
- 05 Results are returned as a DataFrame in the dashboard

SQL Generation Pipeline

Controlled LLM workflow

- 01 1. Capture user question
- 02 2. Inject known database schema
- 03 3. Ask Qwen3 for ONE SELECT query
- 04 4. Remove markdown wrappers
- 05 5. Validate the query
- 06 6. Execute only approved SQL
- 07 7. Display results

Security & Safety

Read-only by design

- Blocked: INSERT

- Blocked: DELETE

- Blocked: TRUNCATE

- Blocked: MERGE

- Blocked: UPDATE

- Blocked: DROP / ALTER / CREATE

- Blocked: EXEC / EXECUTE

- Production hardening: dedicated SELECT-only database account

Defense in Depth

Application guardrail + database permissions

- 01 LLM prompt restrictions are not a security boundary
- 02 Application validation provides a second control
- 03 A dedicated SELECT-only DB identity provides a stronger final boundary
- 04 Principle: assume generated output can be wrong and constrain execution

SQL Server Monitoring

Know the database state before acting

- 01 Connection health check through PyODBC
- 02 @@SERVERNAME identifies the server
- 03 DB_NAME() confirms database context
- 04 GETDATE() provides SQL Server time
- 05 Dashboard exposes ONLINE / OFFLINE state
- 06 Connection failures are surfaced with error details

WhatsApp Alerting

Turn infrastructure state into an actionable message

- 01 Health check → alert payload → WhatsApp Cloud API
- 02 Recipient and credentials are environment-configured
- 03 Alert includes server, database, time and status
- 04 REST API integration uses authenticated POST requests
- 05 Current dashboard supports manual alert sending

Streamlit Operations Dashboard

A single operational surface

- 01 AI SQL Assistant
- 02 SQL Server Health Monitor
- 03 WhatsApp Alert Center
- 04 Generated SQL viewer
- 05 Query results DataFrame
- 06 Connection / model / WhatsApp status
- 07 Error handling and environment-based configuration

Technology Stack

Open, practical and portfolio-ready

- Python — application logic

- LangChain — LLM integration

- Pandas — result handling

- WhatsApp Cloud API + REST — notifications

- Qwen3 4B + Ollama — local LLM

- Microsoft SQL Server + PyODBC — database

- Streamlit — UI

- LangSmith — planned observability expansion

Configuration & Secrets

Keep credentials out of source code

- 01 SQL and API settings are loaded from `.env`
- 02 Access tokens are not hard-coded
- 03 Database and messaging configuration are environment-driven
- 04 Git hygiene: `.env` excluded through `.gitignore`
- 05 Production next step: managed secrets / vault-based configuration

Engineering Workflow

Four weeks of iterative delivery

- 01 Week-long cycles of learning and implementation
- 02 Coding → testing → debugging → refinement
- 03 API integration troubleshooting
- 04 Database connectivity validation
- 05 UI integration and usability improvements
- 06 Architecture refined toward production-oriented practices

Learning & Engineering Outcomes

More than an LLM demo

- SQL Server connectivity and Python database integration

- Local LLM deployment with Ollama

- REST API integration

- Streamlit development

- Natural-language-to-SQL generation

- LangChain integration

- WhatsApp API integration

- Error handling, debugging and architecture

Reliability & Production Thinking

What makes the project enterprise-oriented

- 01 Visible health status before operations
- 02 Read-only execution model
- 03 Explicit validation and failure handling
- 04 Secrets separated from code
- 05 Modular services for monitoring and messaging
- 06 Transparent generated SQL
- 07 Clear path to authentication, audit and observability

Next-Generation Roadmap

From portfolio project to production platform

- Automatic SQL Server service-start detection

- LangSmith tracing and evaluation

- Authentication & authorization

- AI-generated SQL evaluation

- Cloud deployment

- Automatic WhatsApp notifications

- Audit logging

- Docker + CI/CD

- Production-grade DB permissions

Career Value & Interview Story

A bridge from SQL administration to AI engineering

- 01 Combines database expertise with modern GenAI application engineering
- 02 Demonstrates end-to-end thinking: model → system → product
- 03 Shows practical API, UI, security and monitoring integration
- 04 Designed as an interview-discussion project with explainable architecture
- 05 Target roles: ML Engineer • Data Science & ML • GenAI Engineer • AI Application Developer

PROJECT COMPLETED — NEXT BUILD BEGINS

Four weeks. One integrated platform. A stronger engineering foundation.

- 01 Built with consistent effort, learning, debugging and refinement
- 02 GitHub: github.com/Taqhi-adf/taqhi_ai.git
- 03 LinkedIn: www.linkedin.com/in/taqhi
- 04 Email: taqhibaba9@gmail.com | taqhibaba11@gmail.com
- 05 Thank you — ready to discuss architecture, implementation and design decisions